STAT 579: Applied Causal Inference

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Framed as a Target Trial Emulation

Purpose

The final project is designed to integrate **conceptual, methodological, and applied skills** in causal inference. Students will **pose and answer a causal question** using real or simulated data, grounding their analysis in the **target trial emulation framework**.

The project should demonstrate:

- Clear articulation of a **STAT 579 Final Project Guidelinesusal estimand** (not just a statistical parameter).
- Careful **identification strategy** with explicit assumptions.
- Implementation of at least one **modern estimation method** (e.g., IPW, matching, IV, MSM).
- **Critical sensitivity checks** and interpretation within a real-world context.

Timeline & Deliverables

1. **Week 1–2: Causal Question Definition**
   - Submit a 1–2 paragraph causal question relevant to **biostatistics, epidemiology, or social/biomedical sciences**.
   - Include justification of why the question is causal (not associational).
2. **Week 3–4: Preliminary DAG**
   - Provide a directed acyclic graph (using `dagitty` or `ggdag`).
   - State which paths must be blocked for identification.
   - Identify potential sources of bias (confounding, selection, measurement error).

3. **Week 5–6: Identification Strategy**
   - Explicitly state whether exchangeability, positivity, SUTVA, and consistency are plausible.
   - Describe your strategy: backdoor adjustment, IV, MSM, or mediation.
4. **Week 9: Project Proposal (10% of course grade)**
   - 3–5 pages.
   - Elements:
     - Research question & target trial specification (eligibility, treatment strategies, follow-up, outcome).
     - DAG & assumptions.
     - Identification strategy.
     - Planned data source (public dataset, your own research, or simulation).
     - Anticipated challenges.
5. **Weeks 10–13: Analysis**
   - Implement at least one causal method (e.g., IPW, g-computation, IV, mediation, MSM).
   - Provide R code and output with clear interpretation.
   - Conduct diagnostics (balance checks, overlap, weak instrument tests, model fit).
   - Perform at least one **sensitivity analysis** (e.g., E-values, Rosenbaum bounds, tipr).
6. **Week 14: Presentation (10%)**
   - 12–15 minutes, conference-style.
   - Emphasize assumptions, robustness, and substantive conclusions—not just technical detail.
7. **Final Paper (25%) – Due Finals Week**
   - ~12–15 pages, structured as a research article:
     1. **Introduction**: Causal question and motivation.
     2. **Target Trial Protocol**: Eligibility, treatment, follow-up, outcome, causal estimand.
     3. **Assumptions & Identification**: DAG, assumptions, strategy.
     4. **Methods**: Data source, causal methods used, software.
     5. **Results**: Estimates, sensitivity analyses, robustness checks.
     6. **Discussion**: Interpretation, limitations, implications.
     7. **Appendix**: R code (clean, annotated).

Detailed Instructions

Step 1: Choosing a Question

- Prefer **health, social, or biomedical contexts** (e.g., effect of smoking on lung cancer, impact of job training on income, role of diet on heart disease).
- The question must be **causal**, not descriptive.
  - *Bad*: "What factors are associated with depression?"
  - *Good*: "What is the effect of exercise on reducing depression severity?"

Step 2: DAG Construction

- Must use `dagitty` or `ggdag`.
- Include **confounders, colliders, mediators, and selection mechanisms**.
- Explicitly identify adjustment set(s) and whether positivity holds.

Step 3: Identification & Assumptions

- Address the **three pillars**:
  1. **Exchangeability** (unconfoundedness, or IV/exclusion restriction justification).
  2. **Positivity** (overlap diagnostics, trimming if needed).
  3. **Consistency & SUTVA** (intervention is well-defined).

Step 4: Method Implementation

- Choose a method appropriate to your design:
  - **Backdoor adjustment**: matching, stratification, regression, IPW.
  - **IV**: LATE, 2SLS.
  - **Mediation**: NDE/NIE decomposition.
  - **MSM**: time-varying exposures.
- Use R packages: `MatchIt`, `WeightIt`, `ivreg`, `mediation`, `EValue`, `tipr`.

Step 5: Sensitivity Analysis

- Required to evaluate robustness. Options:
  - E-value.
  - Rosenbaum bounds.
  - Simulated confounding (tipr).

Step 6: Writing & Presentation

- Clarity of assumptions is more important than fancy methods.
- Discussion must explicitly state: *"If assumption X fails, my estimate is biased because..."*

Evaluation Criteria

- **Question & Relevance (10%)**: Is the question causal and well-motivated?
- **DAG & Identification (20%)**: Correct, complete, and clearly explained.
- **Methodology (20%)**: Appropriate method, correct implementation, diagnostics.
- **Sensitivity & Robustness (15%)**: At least one sensitivity analysis, well-interpreted.
- **Results & Interpretation (15%)**: Clear, correct, cautious interpretation.
- **Writing & Presentation (20%)**: Organization, clarity, reproducibility.

References for Guidance

- Hernán & Robins (2024). *Causal Inference: What If*.
- Neal (2020). *Introduction to Causal Inference*.
- Imbens & Rubin (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences* oai_citation:2‡Imbens Rubin 2015 Causal Inference for Statistics Social and Biomedical Sciences An Introduction.pdf.
- Brumback (2022). *Fundamentals of Causal Inference with R* oai_citation:3‡Brumback 2021 Fundamentals of Causal Inference with R.pdf.
- VanderWeele (2015). *Explanation in Causal Inference*.